INTEGRAL CALCULUS CHEAT SHEET

Quick Reference Guide: Definite, Double, & Triple Integrals

1. DEFINITE INTEGRALS

Fundamental Theorem of Calculus

If $f(x)$ is a continuous function on $[a,b]$ and $\int f(x)dx = g(x) + c$, then:

$$\int_a^b f(x)dx = g(x)\Big|_a^b = g(b) - g(a)$$

General Properties

1. $\int_a^b f(x)dx = \int_a^b f(t)dt$
2. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
3. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$
4. $\int_0^a f(x)dx = \int_0^a f(a-x)dx$
5. $\int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$
6. $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

2. LEIBNITZ THEOREM

Differentiation Under Integral Sign

If $\phi(x)$ and $\psi(x)$ are defined and differentiable, and $f(x,t)$ is continuous:

$$\frac{d}{dx}\left[\int_{\phi(x)}^{\psi(x)} f(x,t)dt\right] = \int_{\phi(x)}^{\psi(x)} \frac{\partial f(x,t)}{\partial x}dt + \frac{d}{dx}\{\psi(x)\}f(x,\psi(x)) - \frac{d}{dx}\{\phi(x)\}f(x,\phi(x))$$

Special Case (Function of t only):

$$\frac{d}{dx}\left[\int_{\phi(x)}^{\psi(x)} f(t)dt\right] = \frac{d}{dx}\{\psi(x)\}f(\psi(x)) - \frac{d}{dx}\{\phi(x)\}f(\phi(x))$$

3. DOUBLE INTEGRALS

Definition

Double integral of $f(x,y)$ over a region is denoted by:

$$\iint f(x,y)dxdy \quad \text{or} \quad \iint f(x,y)dydx$$

Transformation of Variables (Jacobian)

To solve double integrals by transforming variables (e.g. to u, v):

$$dxdy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dudv = |J|dudv$$

Transformation in Polar Form

Let $x = r\cos\theta, y = r\sin\theta$. Then the Jacobian is r:

$$dxdy = \frac{\partial(x,y)}{\partial(r,\theta)}drd\theta = r\,drd\theta$$

Area & Volume

- Area of region D: $A = \iint_D dxdy$
- Volume of solid region: $V = \iint z\,dxdy$

4. CURVE TRACING

Cartesian Curves:

- **Symmetry:** If the power of y is even, curve is symmetric about the x-axis. If the power of x is even, symmetric about the y-axis.
- **Origin:** If $f(0,0) = 0$, the curve passes through the origin.

Polar Curves:

- **Symmetry:** If $f(r,-\theta) = f(r,\theta)$, it is symmetric about the initial line ($\theta = 0$).
- **Pole:** Put $r = 0$, then find the value of θ (gives tangent at pole).

5. TRIPLE INTEGRALS

Evaluation
Integral of a function with respect to three variables x, y, z:

$$\iiint f(x,y,z)dxdydz$$

Volume
Volume of a solid region by triple integration:

$$V = \iiint dV = \iiint dxdydz$$

Dirichlet's Theorem
Subject to condition $x + y + z \le 1$ and all variables positive:

$$\iiint x^{l-1}y^{m-1}z^{n-1}dxdydz = \frac{\Gamma(l)\Gamma(m)\Gamma(n)}{\Gamma(l+m+n+1)}$$

In general, for $x_1 + x_2 + ... + x_n \le 1$:

$$\int \cdots \int x_1^{m_1-1} \ldots dx_n = \frac{\Gamma(m_1)\ldots\Gamma(m_n)}{\Gamma(m_1 + ... + m_n + 1)}$$

6. SOLID OF REVOLUTION

Volume of Solid of Revolution

- About **x-axis** ($y = f(x)$ from $x = a$ to $x = b$):

$$V = \int_a^b \pi y^2 dx$$

- About **y-axis** ($x = f(y)$ from $y = c$ to $y = d$):

$$V = \int_c^d \pi x^2 dy$$

Area of Surface of Revolution

- About **x-axis**:

$$S = \int_a^b 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- About **y-axis**:

$$S = \int_c^d 2\pi x\sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

- **Parametric Form** (Revolution about x-axis):

$$S = \int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- **Polar Form** (About initial line):

$$S = \int_\alpha^\beta 2\pi(r\sin\theta)\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$